

Groundwater Notes

CHAPTER 1: INTRODUCTION TO GROUNDWATER

1.1 Global Water Distribution

Water is the most abundant substance on Earth, covering approximately 71% of the Earth's surface. The breakdown of global water reserves is as follows:

- **Oceans:** 96.5%
- **Freshwater:** 2.5%
- **Other Saline Water:** 0.9%

Of the 2.5% that is Freshwater:

- **Glaciers and Ice Caps:** 68.7%
- **Groundwater:** 30.1%
- **Surface and Other Freshwater:** 1.2% (includes lakes, soil moisture, atmosphere, and rivers)

1.2 Definition of Groundwater

Groundwater (also known as sub-surface water) is defined as the water present beneath the Earth's surface in the pore spaces of soil and in the fractures of rock formations. It exists in the zone of saturation.

1.3 Classification of Groundwater (Based on Origin)

Groundwater is classified into the following types based on its source:

- **Meteoric Water:** Derived from atmospheric precipitation (rain and snow). Generally, this water originates from infiltration.
- **Connate Water:** Water entrapped in the interstices (pores) of sedimentary rock at the time of its deposition. It is often highly saline due to long contact with minerals.
- **Juvenile Water (Magmatic Water):** Derived from the interior of the Earth, typically associated with magma or volcanic activity. It is formed from the condensation of steam and magmatic fluids entrapped in rock pores.
- **Metamorphic Water:** Water that is squeezed or drained out of rocks during the process of metamorphism.
- **Fossil Water:** A form of connate water that has been preserved in an aquifer for a very long period of time.

1.4 The Hydrological Cycle

The Hydrological Cycle is the continuous storage and movement of water between the biosphere, atmosphere, and lithosphere. Throughout this process, water changes between different states, such as evaporation to water vapor, water freezing into ice, and water vapor condensing to water.

Key Components of the Cycle:

- Evaporation
- Transpiration
- Sublimation
- Condensation
- Precipitation
- Runoff
- Infiltration and Percolation
- Groundwater Flow

1.5 Surface Water vs. Groundwater

Both surface water and sub-surface water are related through the hydrological cycle (continuous circulation of water).

Comparison of Characteristics:

- **Availability & Location**
 - *Surface Water*: Available on the surface in the form of oceans, seas, and ponds.
 - *Groundwater*: Located below the earth's surface.
- **Contamination**
 - *Surface Water*: Generally more contaminated.
 - *Groundwater*: Less contaminated generally.
- **Mineral Content**
 - *Surface Water*: Low in minerals; considered as soft water.
 - *Groundwater*: High in minerals; considered as hard water.
- **Primary Uses**
 - *Surface Water*: Uses include drinking (usually requires treatment), irrigation, and daily life.
 - *Groundwater*: Uses include domestic supply, industrial purpose, and irrigation.

HYDROGEOLOGICAL FORMATIONS

To know the flow path of groundwater (GW), identify recharge sources, and outflow boundaries, we must understand the underground flow patterns. Generally, there are distinct types of geological formations/layers that determine the availability of GW sources: namely Aquifer, Aquitard, Aquiclude, and Aquifuge.

1. Aquifer

- **Definition:** A formation that is highly porous and permeable.
- **Function:** It not only stores but provides an adequate and significant source of Groundwater (GW).
- **Examples:** Limestone, gravel, fractured basalt.

2. Aquitard

- **Definition:** A formation with partially low porosity.
- **Example:** Sandy clay.

(Note: The list in the image includes Aquiclude and Aquifuge, but their specific definitions were not written out in detail on this page).

CHARACTERISTICS OF GROUNDWATER

General Characteristics

- **Movement:** Groundwater includes slow movement and has a stable temperature.
- **Natural Filtration:** Groundwater is generally purer than surface water.
- **Quality Variations:**
 - Quality varies with local geology and potential contaminants.
 - Dissolved minerals, pH, and hardness vary significantly by location.

Age of Groundwater

- **Definition:** The time since water entered an aquifer.
- **Range:** Ranges from days to years.
- **Why Age Matters?**
 - To identify the pollution of GW.
 - To understand how quickly it moves through aquifers.
 - To identify recharge zones.
- **Measurement:** Scientists use natural tracers to measure age, such as:
 - Carbon-14 (C-14)
 - Chlorofluorocarbons (CFCs)
 - Tritium
 - Sulfur

SPRINGS

Definition

A spring is an outlet for groundwater stored in the rock which is released onto the surface at points where the water table meets the surface.

Types of Springs

1. Tilted Spring

- *Formation:* Occurs where permeable and impermeable rocks alternate.
- *Mechanism:* Water emerges at the base of permeable rocks. Water may percolate downwards until it reaches a rock joint which emerges at the surface.
- *Result:* Causes the water table to reach the surface.

2. Dyke Spring

- *Formation:* Involves a dyke (impermeable rock intrusion).
- *Mechanism:* The dyke acts as a barrier to the flow, forcing water to surface.

3. Karst Spring

- *Location:* Limestone regions.
- *Composition:* Involves calcium carbonate rock.
- *Structure:* Often associated with underground tunnels.

HYDROGEOLOGICAL FORMATIONS (CONTINUED)

• Aquiclude

- *Definition:* An impermeable layer that has good storage capacity and low transmitting capacity.
- *Function:* It generally acts as a barrier to groundwater flow.
- *Example:* Clay.
- *Note:* It allows water through it but doesn't provide enough available water.

• Aquifuge

- *Definition:* A relatively impermeable formation that neither holds (contains) nor transmits water.
- *Characteristics:* It has no interconnected pores.
- *Examples:* Granite, Basalt.

TYPES OF AQUIFERS

Groundwater aquifers are classified into four main types based on their geological confinement:

1. Unconfined Aquifer

- *Definition:* An aquifer that has no overlying impermeable layer and is open to the atmosphere.
- *Recharge:* Recharged directly through the surface by precipitation.
- *Upper Boundary:* The unconfined upper boundary is the water table.
- *Dynamics:* The water level can rise and fall based on rainfall.

- *Pressure:* Water is at atmospheric pressure.

2. **Confined Aquifer (Artesian Aquifer)**

- *Definition:* An aquifer that is sandwiched between two impermeable layers.
- *Pressure:* Water is usually under pressure. It is also known as a pressure aquifer or artesian aquifer.
- *Characteristics:* It is completely filled with water and has no free water table.
- *Flow:* Without pumping, the pressure can cause water to rise naturally.

3. **Leaky Aquifer (Semi-Confined)**

- *Definition:* An aquifer bounded by one or two aquitards.
- *Mechanism:* It allows vertical water movement or leakage through the bounding layers.

4. **Perched Aquifer**

- *Definition:* A special type of unconfined aquifer that occurs above the regional water table.
- *Formation:* It sits on a localized impermeable layer (aeration zone saturation) separate from the main aquifer below.

PROPERTIES OF SOIL/ROCK AFFECTING GROUNDWATER FLOW

The flow of groundwater is heavily influenced by the properties of the soil and rock formations:

- **Porosity:**
 - The total number of voids (empty spaces) in the material.
 - It defines the storage capacity of the formation.
- **Permeability:**
 - The ability of the material to transmit fluids.
 - *Comparison:*
 - *Clay:* Low permeability, but high porosity (holds water but doesn't let it flow).
 - *Sand:* High permeability.
- **Geological Structures:**
 - Features such as rock structures, faults, and fractures significantly impact flow paths.
- **Texture & Grain Size:**
 - *Coarse materials:* Highly permeable and have better water-holding capacity than silt and clay.
- **Specific Yield:**
 - The actual volume of water that drains out of a saturated rock due to gravity.
- **Rock Joint Fractures:**
 - Fractures in the rock that serve as conduits for water movement.

CHAPTER 5: AQUIFER PROPERTIES (CONTINUED)

5.1 Specific Yield (S_y)

• **Definition:** Specific yield is the actual volume of water that drains out of a saturated rock or soil due to gravity.

• **Formula:**

$$S_y = \frac{V_w}{V_t}$$

Where:

- V_w = Volume of water drained
- V_t = Bulk volume of soil or rock

Typical Values of Specific Yield:

- Gravel: 24% (Medium size)
- Sand: 28% (Medium size)
- Silt: 8%
- Clay: 3%
- Limestone: 14%

Example Problem 1: Estimating Drawdown

Question: Estimate the average drawdown over an area where 25 million m^3 of water has been pumped through a number of uniformly distributed wells. The area is $250km^2$ and the specific yield of the unconfined aquifer is 25%.

Solution:

Given:

- Volume of water drained (V_w) = $25 \times 10^6 m^3$
- Area (A) = $250km^2 = 250 \times 10^6 m^2$
- Specific Yield (S_y) = 25% = 0.25

Calculation: Using the formula:

$$S_y = \frac{V_w}{V_{total}} = \frac{V_w}{Area \times \text{drawdown}(h)}$$

Rearranging for h :

$$\begin{aligned} h &= \frac{V_w}{Area \times S_y} \\ h &= \frac{25 \times 10^6}{250 \times 10^6 \times 0.25} \\ h &= \frac{25}{62.5} = 0.4m \end{aligned}$$

(Note: Your handwritten note shows a calculation resulting in 0.67m, but based on the numbers written: $25/(250 \times 0.25) = 25/62.5 = 0.4$. Please check if the area was meant to be $150km^2$ as per the scribble in the calculation line).

CHAPTER 6: FUNDAMENTALS OF GROUNDWATER MOTION

6.1 Darcy's Law

Definition: An empirical law formulated by Henry Darcy which describes the flow of fluid through a porous medium.

Statement: The flow rate (Q) through a porous media is proportional to the cross-sectional area (A) and the head loss (h_L) inversely proportional to the length (L).

Formula:

$$Q = -KA \frac{dh}{dL} \quad \text{or} \quad Q = KAi$$

Where:

- Q = Flow rate (m^3/s)
- K = Coefficient of Permeability (Hydraulic Conductivity)
- A = Cross-sectional area of the pipe/aquifer
- $\frac{dh}{dL}$ or i = Hydraulic gradient
- **Negative sign:** Indicates flow of water in the direction of decreasing head.

6.2 Hydraulic Conductivity (K)

- **Definition (Metric Units):** Defined as the rate of flow in cubic meters per day through a cross-sectional area of $1m^2$ under a hydraulic gradient of $1m/m$.
- **Factors:** Change in temperature affects hydraulic conductivity because it changes the viscosity of the water.

Example Problem 2: Determining Hydraulic Conductivity

Question: A field sample of an unconfined aquifer is packed in a cylinder. The length and diameter of the cylinder are 50 cm and 6 cm respectively. The field sample is tested for a period of 3 minutes under a constant head difference of 16.3 cm. As a result, $45.2cm^3$ of water is collected at the outlet. Determine the hydraulic conductivity of the aquifer sample.

Solution:

Given:

- Length (L) = 50 cm
- Diameter (D) = 6 cm $\rightarrow$ Radius (r) = 3 cm
- Time (t) = 3 min = 180 sec
- Head difference (h) = 16.3 cm
- Volume (Vol) = $45.2cm^3$

Step 1: Calculate Discharge (Q)

$$Q = \frac{Volume}{Time} = \frac{45.2}{180} = 0.251 \text{ cm}^3/s$$

Step 2: Calculate Area (A)

$$A = \pi r^2 = \pi \times (3)^2 = 28.27 \text{ cm}^2$$

Step 3: Calculate Gradient (i)

$$i = \frac{h}{L} = \frac{16.3}{50} = 0.326$$

Step 4: Calculate K

$$Q = K A i \rightarrow K = \frac{Q}{A i}$$

$$K = \frac{0.251}{28.27 \times 0.326}$$

$$K = 0.0272 \text{ cm/s}$$

$$(or \approx 23.5 \text{ m/day})$$

CHAPTER 7: FLOW VELOCITIES & VALIDITY

7.1 Darcy Velocity (v)

- Also known as Discharge Velocity.
- It is referred to as Darcy's velocity because it assumes that flow occurs through the entire cross-section of the material without regard to solids and pores.
- **Formula:**

$$v = \frac{Q}{A} = -K \frac{dh}{dL}$$

7.2 Seepage Velocity (v_a)

- Also known as Average Interstitial Velocity.
- This is the actual velocity limited only to the pore spaces.
- **Formula:**

$$v_a = \frac{v}{\alpha}$$

Where α = Effective porosity.

7.3 Validity of Darcy's Law

- Darcy's Law is valid only in Laminar Flow.
- The Reynolds number (Re) should be less than 1.

CHAPTER 8: LIMITATIONS & VALUES

8.1 Limitations of Darcy's Law

Darcy's law is not a universal law like Newton's laws. It has specific limitations:

- **Laminar Flow Only:** It is only valid for laminar flow regimes where the Reynolds number (Re) is typically less than 1 (some sources say 1-10).
- **Newtonian Fluids:** It is not applicable for Non-Newtonian fluids.
- **Application:** It is most common in groundwater flow because flow velocities there are generally very slow.

8.2 Hydraulic Conductivity (K) of Various Materials

Typical values for hydraulic conductivity (K) based on material type:

Material Type	Hydraulic Conductivity (K)
Medium Gravel	270 m/day
Sand (Medium)	12 m/day
Silt	0.08 m/day
Clay	0.0002 m/day
Limestone	0.94 m/day
Basalt	0.01 m/day

(Note: The upper limit of validity for Darcy's law can be extended up to $Re = 10$ in some contexts).

CHAPTER 9: NUMERICAL PROBLEMS (ADVANCED)

Problem 3: Validity Check (Reynolds Number)

Question: The sample has a median grain size (d_{50}) of 0.037 cm and a porosity of 0.3. The test is conducted using pure water at 20°C. Determine the Darcy velocity, Average Interstitial velocity, and assess the validity of Darcy's Law.

Your Solution (from notes):

1. Calculate Darcy Velocity (v): Using values from the previous problem ($K = 23.5$ m/day, $i = 0.326$):

$$v = K \times \frac{h}{L}$$
$$v = 23.5 \times 0.326 = 7.661 \text{ m/day}$$

2. Calculate Average Interstitial Velocity (v_{avg}):

$$v_{avg} = \frac{v}{\alpha} = \frac{7.661}{0.3}$$
$$v_{avg} = 25.54 \text{ m/day}$$

3. Check Validity (Re):

$$Re = \frac{vd}{\nu} \quad \text{or} \quad \frac{\rho vd}{\mu}$$

(Note: You marked this calculation as "Maybe wrong")

$$Re = \frac{25.54 \times 0.037}{86400 \times 10^{-5}} = 1.09$$

Conclusion: $1.09 \approx 1$ (Valid)

CORRECTION CHECK (Why it might be wrong): The issue in the calculation is likely Unit Consistency.

- v_{Darcy} should be used for Reynolds number, not v_{avg} (usually).
- Velocity is in m/day, but diameter is in cm, and viscosity is usually in cm^2/s or m^2/s .

Corrected Units:

- $v = 7.66 \text{ m/day} = 0.0088 \text{ cm/s}$
- $d = 0.037 \text{ cm}$

- ν (water at 20°C) $\approx 0.01 \text{ cm}^2/\text{s}$
- $Re = \frac{vd}{\nu} = \frac{0.0088 \times 0.037}{0.01} \approx 0.033$

Since $0.033 < 1$, Darcy's Law is definitely Valid.

Problem 4: Variable K and Thickness

Question: A confined aquifer with a horizontal bed has a varying thickness as shown in the figure. The aquifer is inhomogeneous with hydraulic conductivity $K = 12 + 0.006x$. Piezometric heads at section 1 ($x = 0$) and section 2 are 14.2 m and 18.8 m respectively. Measured above the upper confining layer. Assume the flow in the aquifer is essentially horizontal. Determine the flow rate per unit width.

Given Data:

- Hydraulic Conductivity: $K(x) = 12 + 0.006x$
- Head at x_1 : $h_1 = 14.2\text{m}$
- Head at x_2 : $h_2 = 18.8\text{m}$
- Aquifer Thickness: $b(x)$ (Equation cut off in image, looks like $30 + \dots$)
- Area (A): $A = b(x) \times 1$ (unit width)

Solution Setup: Since K and Area (A) both change with distance x , we must integrate Darcy's Law:

$$Q = -K(x)A(x)\frac{dh}{dx}$$

$$Qdx = -(12 + 0.006x)(b(x))dh$$

(You would then integrate both sides from x_1 to x_2 to solve for Q .)

Problem 4: Variable K and Thickness (Solution Continued)

Given Data:

- Length (L): 3600 m
- Head 1 (h_1): 14.2 m (at $x = 0$)
- Head 2 (h_2): 18.8 m (at $x = 3600$)
- Hydraulic Conductivity (K): varies linearly as $K(x) = 12 + 0.006x$.

Step 1: Determine Aquifer Thickness Equation $b(x)$ The thickness varies linearly from 30m (at $x = 0$) to 75m (at $x = 3600$). Using the straight-line equation $y = mx + c$:

$$b(x) = 30 + \frac{75 - 30}{3600}x$$

$$b(x) = 30 + 0.0125x$$

Step 2: Determine Area (A) Since we are calculating for unit width:

$$A(x) = b(x) \times 1 = 30 + 0.0125x$$

Step 3: Integrate Darcy's Law Since K and A are functions of x , we cannot just multiply. We must integrate:

$$Q = -K(x)A(x)\frac{dh}{dx}$$

Rearranging to separate variables:

$$\frac{Q}{K(x)A(x)}dx = -dh$$

$$\int_0^{3600} \frac{dx}{(12 + 0.006x)(30 + 0.0125x)} = -\frac{1}{Q} \int_{14.2}^{18.8} dh$$

Step 4: Solve the Integral (From your notes, the definite integral of the left side yields approximately 3.7776)

$$3.7776 = -\frac{1}{Q}[18.8 - 14.2]$$

$$3.7776 = \frac{4.6}{Q}$$

$$Q = \frac{4.6}{3.7776}$$

Flow Rate (Q) = 1.2 m²/day

CHAPTER 10: INTRINSIC PERMEABILITY (K_I)

10.1 Definition

Intrinsic Permeability (k_i) is the property of a porous medium that characterizes its ability to transmit a fluid.

- It is dependent only on the physical properties of the porous medium itself, such as:
 - Grain size
 - Grain shape and arrangement
 - Pore interconnection

10.2 Difference from Hydraulic Conductivity (K)

While Intrinsic Permeability (k_i) depends only on the medium (rock/soil), Hydraulic Conductivity (K) depends on the properties of both the porous media and the fluid (viscosity and density).

10.3 The Formula

The relationship between Intrinsic Permeability and Hydraulic Conductivity is:

$$K = k_i \frac{\rho g}{\mu}$$

Or rearranged to find k_i :

$$k_i = \frac{K\mu}{\rho g}$$

Where:

- K = Hydraulic Conductivity (m/s)
- μ = Dynamic Viscosity of the fluid ($kg/m \cdot s$)
- ρ = Density of the fluid (kg/m^3)
- g = Acceleration due to gravity (m/s^2)

Units: Intrinsic Permeability is measured in Darcy or m^2 .

CHAPTER 11: NON-DARCYIAN FLOW

11.1 Definition

Non-Darcyian Flow refers to flow regimes where the linear relationship between hydraulic gradient (i) and flow rate (q) (Darcy's Law) deviates. This typically happens when flow becomes turbulent.

11.2 Causes of Deviation

Typical Darcy flow is laminar. Deviation occurs due to:

- **Turbulent Effect:** High velocities cause eddies and turbulence.
- **Inertial Effect:** At high speeds, water cannot negotiate tight pore turns smoothly.
- **Heterogeneous Effect:** Complex flow paths in fractured rocks.

11.3 Examples of Non-Darcyian Flow

- **Near Well Bores:** Water flowing rapidly towards a well bore during pumping often exceeds laminar conditions and enters turbulent conditions.
- **Oil & Gas Extraction:** Gas flows rapidly towards the extraction point.
- **Fractured Rock:** Water flowing through large complex fractures or karst conduits.

CHAPTER 12: DETERMINATION OF HYDRAULIC CONDUCTIVITY

There are several methods used to determine the Hydraulic Conductivity (K) of soil or rock:

12.1 List of Methods

- Empirical Methods
- Laboratory Methods (Constant head & Variable head)
- Tracer Test (Travel of tracer between two wells)
- Auger Hole Test (For water at shallow levels)
- Pumping Test of Oil/Water

12.2 Empirical Methods

Various scientists have studied the relationship between hydraulic conductivity and the properties of porous media.

General Formula:

$$K = CD^2$$

Where:

- C = Constant
- D = Grain diameter

Specific Formula: Some specific forms express the formula as:

$$K = F_s F_n d^2$$

Where:

- F_s = Grain shape factor
- F_n = Grain porosity factor

12.3 Laboratory Methods (Permeameters)

In the lab, Hydraulic Conductivity (K) is determined using a device called a Permeameter. Flow is maintained through a small sample of material while measuring flow rate and head loss. There are two main types of permeameters:

A. Constant Head Permeameter

- **Suitability:** Best for Coarse Grain soils (Sand, Gravel) which have high flow rates.
- **Mechanism:** Water enters the medium cylinder from the bottom and is collected as overflow after passing through the sample. The head (pressure) remains constant throughout the test.

Formula:

$$K = \frac{VL}{Ath}$$

Where:

- V = Volume of water collected
- L = Length of sample
- A = Cross-sectional area of sample
- t = Time of measurement
- h = Head loss

B. Variable Head (Falling Head) Permeameter

- **Suitability:** Used for finer soils where flow is slower (implied).
- **Mechanism:** Water is added to a thin "fault tube" (standpipe). As water flows through the sample, the water level in the tube falls over time.

Derivation: The rate of flow is governed by Darcy's Law. Using the continuity equation:

$$Q_{tube} = Q_{sample}$$

$$a \frac{dh}{dt} = -KA \frac{h}{L}$$

(Where ' a ' is the area of the tube and ' A ' is the area of the sample)

Final Formula: By integrating from time 0 to t and head h_1 to h_2 :

$$K = \frac{aL}{At} \ln \left(\frac{h_1}{h_2} \right)$$

or

$$K = 2.303 \frac{aL}{At} \log_{10} \left(\frac{h_1}{h_2} \right)$$

Where:

- a = Area of the standpipe/tube

- h_1 = Initial head
- h_2 = Final head after time t

CHAPTER 12: DETERMINATION OF HYDRAULIC CONDUCTIVITY (CONTINUED)

12.3 B. Variable Head (Falling Head) Permeameter Details

Test Setup & Mechanism:

- **Process:** Water is added to the "fault tube" (standpipe). It flows upward through the cylindrical sample and is collected as an overflow.
- **Measurement:** The test consists of measuring the rate of fall of the water level in the tube and collecting the volume of water overflow through time.

Derivation of the Formula: The rate of flow is governed by Darcy's Law. Using the Continuity Equation:

$$Q_{tube} = Q_{sample}$$

Flow from Tube:

$$Q_{tube} = a \times \frac{dh}{dt}$$

Where:

- a = Area of the tube
- $\frac{dh}{dt}$ = Rate of fall of head in the tube

Flow through Sample (Darcy's Law):

$$Q_{sample} = KiA$$

Where:

- A = Cross-sectional area of the sample
- i = Hydraulic gradient = $\frac{h}{L}$

Equating and Integrating:

$$a \frac{dh}{dt} = -KA \frac{h}{L}$$

(Note: The negative sign indicates the head is decreasing over time)

Rearranging to solve for K :

$$\frac{a \cdot L}{A \cdot K} \int_{h_1}^{h_2} \frac{dh}{h} = - \int_0^t dt$$

Final Formula:

$$K = \frac{aL}{At} \ln \left(\frac{h_1}{h_2} \right)$$

or using log base 10:

$$K = 2.303 \frac{aL}{At} \log_{10} \left(\frac{h_1}{h_2} \right)$$

Suitability (Soil Types):

- **Soil Type:** Fine grain soils with low permeability.
- **Examples:** Silt, Clay.
- **Flow Condition:** Used where there is a low flow rate.
- **Measurement Focus:** The test focuses on measuring the change in head rather than the volume of discharge (which is small).